

Design and testing of a microelectromechanical-system-based high heat flux vaporizing liquid microthruster

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ABSTRACT

This paper presents the design of a **low power (8 W) water-fed vaporizing liquid microthruster** that operates in the **Leidenfrost boiling regime**. The direct thrust measurements show the proposed design is able to produce one of the **best specific impulse amongst all similar microthrusters reported in the literature**. One of the important features that enables the superior performance is the **high heat flux miniature molybdenum heating elements** fabricated inside the vaporization chamber in a **single silicon device package of dimensions 35 mm × 35 mm × 1.05 mm**. The molybdenum heating elements are capable of producing a **heat flux output of 370 kW/m²** resulting in a **sustained operational pressure of 2 bar and temperature of 446 K**. The specific design details, such as choice of configurations and sizing, are given in this paper. In particular, the choice of material and the fabrication process carried out at the in-house nanosystem fabrication facility are described. The experimental setup and uncertainty analysis of the mN-range hanging pendulum direct thrust measurement stand are presented. Then, the **evaporation of liquid water propellant is investigated in detail**. The occurrence of Leidenfrost effect inside the vaporization chamber is captured with a high-speed camera at a frame rate of 5000 Hz. It can be seen that the rapid vaporization of propellant in the Leidenfrost boiling regime results in a high specific impulse amongst other microelectromechanical-system-scale vaporizing liquid microthrusters. The direct thrust measurements show the current design is capable of producing a **specific impulse ranging between 25.9–51.1 s at a continuous thrust level of 1.00–2.00 mN**; considering contemporary energy storage technologies, the estimated thrust per device dry mass and specific impulse per device dry mass levels are **4.0–8.0 mN/kg** and **103.6–204.0 s/kg** respectively.

1. Introduction

The aim of this paper is to propose a **new liquid water-fed resistojet micropropulsion design that could find applications in nanosatellite engineering**. The development of vaporizing liquid thrusters, a type of resistojet, has started since the mid-1960s [1–7]. Sutherland and Maes [8] reviewed the **potential advantages** with the resistojet architecture: **high reliability and durability, low complexity, low inert part weight, competitive thrust-to-weight ratio, and its relatively high thermal efficiency (65%–95% depending on the operating temperature [9])** compared with other propulsion technologies such as cold gas, monopropellant, bi-propellants. High-power (kW-range) resistojets gained popularity in commercial applications in the early 1980's for station keeping and attitude control [10].

Recognizing the **need for low power, low electromagnetic interference propulsion system**, recent reviews reveal that microelectromechanical

system (MEMS) based vaporizing liquid microthrusters, a type of resistojet, are indeed suitable for contemporary low earth orbit satellites [11–14]; the main types of resistojets includes gas-fed resistojet, vaporizing liquid microthruster, and free-molecular micro-resistojets. **Resistojet requires only inert propellant**, making it suitable for use in propulsion system of secondary satellites where the storage of chemical energy is restricted by their respective specifications (as stated in CubeSat and PocketQube design specifications [15,16]). **The flexibility in the use of water and other inert propellants with resistojets eliminates the safety and environmental concerns in the handling of fuel for chemical propulsion**. Liquid water propellant is chosen in the current study as it is storable and chemically inert to **polymeric storage tank materials in flight conditions [17]**, and non-toxic to human and environment; further arguments for using water can be found in reference [18].

A disruptive innovation in resistojet design was brought about by the advent of fabrication techniques for MEMS since the late 1990's.

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Nomenclature

Symbols

$C_{F-\Delta\bar{x}}$	force-displacement calibration factor [N/m]
$\dot{E}$	microthruster heater power consumption [W]
F_{thrust}	thrust [N]
g_0	standard acceleration due to gravity [m/s ²]
h_{OP}	syringe pump plunger position [m]
i	index of the i th variable for uncertainty analysis [-]
I_{ht}	microthruster heater electrical current [A]
I_{sp}	specific impulse [s]
$\dot{m}_{prop}$	propellant mass flow rate [kg/s]
m_{TCh}	mass of propellant accumulated in test chamber [kg]
N_v	total number of variables [-]
P_{prop}	propellant feed pressure [bar]
ΔP_{TCh}	pressure rise in the test chamber [Pa]
R_{ht}	microthruster heater electrical resistance [Ω]
R_{H_2O}	specific gas constant of water vapour [J/kg/K]
$R_{F-\dot{E}}$	thrust-to-power ratio [N/W]
$\dot{S}_{gen}$	entropy generation rate [W/K]
T_g	gas temperature [K]
T_{in}	propellant inlet temperature [K]
T_{out}	heat rejection temperature [K]
T_{prop}	propellant temperature in microthruster [K]

T_{sat}	saturation temperature [K]
T_{TCh}	background temperature in test chamber [K]
ΔT_{bx}	boiling excess temperature [K]
u_c	standard combined uncertainty [-]
$u(x_i)$	uncertainty of the i th variable [-]
V_{ht}	microthruster heater voltage [V]
V_{TCh}	volume of test chamber [m ³]
$\dot{W}_{lost}$	power lost due to entropy generation in evaporation chamber [W]
$\Delta\bar{x}$	time averaged displacement of the pendulum arm [m]
α_{th-exp}	linear thermal expansion coefficient [K ⁻¹]
$\partial f / \partial x_i$	sensitivity in uncertainty contribution with respect to the i th variable [-]

Abbreviations

AWG	American wire gauge
DRIE	deep reactive-ion etching
FMMR	free-molecular micro-resistojets
ICP	inductively coupled plasma
LPCVD	low pressure chemical vapour deposition
MEMS	microelectromechanical system
PECVD	plasma enhanced chemical vapour deposition
VLM	vaporizing liquid microthruster

The work by DeGroot reviewed that some design requirements for resistojets, such as compactness of the heater elements and their installation, could be realized through MEMS-based fabrication techniques [19,20]. The pioneering works [21–31] have explored MEMS-based fabrication technologies for the miniaturization of resistojets.

Over the last 20 years various research workers had started engaging in thruster head design for vaporizing liquid microthrusters based on MEMS fabrication techniques and thrust performance testing. In most design studies, the scope of design studies includes: 1) vaporization chamber; 2) ohmic heater elements; and 3) converging-diverging nozzle, although there are a few examples tested the performance of the vaporization chamber and converging-diverging nozzle without the ohmic heater elements. Notable early examples include [32–35], where these designs have ohmic heaters fabricated as an integral part of the vaporizing liquid microthrusters. Both vaporizing liquid microthrusters from references [32,35] have been tested in a direct thrust measurement stand; whereas the design from references [33,34] was tested in an indirect thrust measurement experiment.

The difference between the direct and indirect thrust measurement methods is that in a direct thrust measurement the thruster is mounted on the thrust stand, the thruster pushes the part of the thrust stand directly; in an indirect thrust measurement the thruster is mounted on a stationary frame separate from the thrust stand, the thruster exhaust plume impinges on a target plate of the thrust stand. It has been pointed out in reference [36] that direct thrust measurement methods are considered more accurate for thrust levels above μ N. For the thrust levels produced by the vaporizing liquid microthrusters reviewed in this section, indirect thrust measurement methods are prone to error associated with the momentum transfer from the exhaust plume to the exhaust plume impingement target plate of the indirect thrust stand [37,38]. The factors affecting the accuracy of indirect thrust measurements mainly are: 1) the distance from thrust exhaust to the impingement target plate, 2) the alignment between the axes of the exhaust plume and that of the impingement target plate, and 3) the condensation of exhaust plume occurring on the impingement target plate.

In general, indirect thrust measurement methods require mechanical setup that is less complex; since then several other groups of research workers had carried out independent work on vaporizing liquid

microthruster design, fabrication and testing based on indirect thrust measurement methods. Notable examples with integral heater element include references [37–42].

There has also been research work on the testing of the vaporization chamber of vaporizing liquid microthrusters in isolation, in this configuration the heating power of which is supplied by heaters external to the MEMS-based devices [43–46]. Vaporizing liquid microthrusters with external heater configuration would lose more heat to the surrounding because the paths of thermal transfer from the heater elements to the propellant become more indirect and thus limiting the thermal efficiency. In the most recent MEMS-based vaporizing liquid microthruster development, reference [47] reports several variants of vaporizing liquid microthruster design with molybdenum heater elements incorporated on the external side of the vaporization chamber; the use of molybdenum enable the temperature of vaporizing liquid microthruster device to be inferred. Note that in Ref. [47] the propulsive performance is not measured.

The thrust characterization methods, propulsion performance, power consumption, and configurations of various MEMS-based vaporizing liquid microthrusters are summarized in Tables 1–3. Note that the specific impulse levels of vaporizing liquid microthrusters estimated using analytical or numerical methods, or measured with indirect thrust measurement techniques tend to be higher than that of the vaporizing liquid microthrusters measured in direct thrust measurement facilities. Amongst MEMS-scale vaporizing liquid microthruster designs reviewed in this section, the highest specific impulse are attained by the designs reported [40,43,47], all of which operates at a temperature above 423 K; unfortunately none of these designs have their performances characterized with a direct thrust measurement method.

The aim of the current work is to realize the potential of the MEMS-based vaporizing liquid microthruster technology to achieve a competitive specific impulse compared with other state-of-the-art resistojets configurations. There are other state-of-the-art resistojets implementations which operate on non-liquid propellants (i.e. gas-fed or solid sublimation) such as: the MEMS-based free-molecular micro-resistojets reported in references [48–50] which has a specific impulse (I_{sp}) of 7–39 s, as well as a recent small-scale Xenon gas-fed resistojets STAR reported in references [51–53] which has a specific impulse of (I_{sp}) of 52.51–80.11 s; both designs were tested with direct thrust measurement

Table 1
MEMS-based vaporizing liquid microthruster comparison: propulsion performance.

Ref.	Thrust data evaluation method		$\dot{m}_{\text{prop}}$ [mg/s]	P_{prop} [bar]	F_{thrust} [μN]	I_{sp} [s]
[32]	direct measurement	force amplified by lever	3.40–8.57	no data	150–460	4.50–5.50
[26]	direct measurement	azimuthal torsional	no data	no data	32	no data
[34]	indirect measurement	cantilever deflection	0.038	no data	0.47 ^a	1.3 ^a
[39]	indirect measurement	cantilever deflection	0.7	no data	120	17.5
[43]	indirect measurement	force amplified by lever	5	2.3	5200	107
[45]	numerical estimation	CFD ^b	2.08–16.67	1.05–1.95	< 6400	no data
[40]	indirect measurement	simulation electronic balance	0.2–2.04	no data	150–1014	50–105
[38]	indirect measurement	piezoresistive sensor	1	1.04	33.6–67.1	4.51–8.15
[47]	theoretical estimation	isentropic nozzle	0.37–3.42	4.80–5.15	750–3860	105.22–115.09
This work	direct measurement	hanging pendulum	3.06–5.99	1.25–2.22	997–2004	25.9–51.1

^a Root mean square value.

^b Computation fluid dynamics.

method and measured mass flow rate. Therefore a major effort has been spent in the current work to ensure the thrust measurements are accurate by the use of direct thrust measurement stand, which is regarded as a more accurate thrust measurement method [36], with the uncertainties are fully quantified according to the industrial standard JCGM 100:2008 [54]. The vaporizing liquid microthruster engineered and fabricated in the current work has been tested in a bespoke mN-range hanging pendulum direct thrust measurement stand, which has a relatively simple design but precisely engineered to hysteresis of less than 0.61% at the range of thrust measurement in the current study. The data set includes thrust and specific impulse measurements as well as propellant pressure and temperature measurements, which makes this work one of the most comprehensive to-date.

This paper documents the engineering, fabrication and test campaign of a MEMS-based vaporizing liquid microthruster in a comprehensive manner, aiming to provide a reference engineering and fabrication method for the popularization of MEMS-based vaporizing liquid microthruster technology. On top of that, this paper provides a detailed investigation on the flow patterns inside the vaporization chamber operating in the film boiling regime in both in the transient and steady states as well as its transient electrical characteristics. These observations should be relevant to all MEMS-based vaporizing liquid microthruster designs equipped with metal heater elements. Overall, the detailed introduction of the vaporizing liquid microthruster design and the discussions of the associated performance shall be considered as the main contribution of this work.

The remainder of this paper is organized as follows: section 2 presents the design strategy for the vaporizing liquid microthruster where the aerothermal design considerations and MEMS-based fabrication constraints are discussed; section 3 presents the detailed design and the MEMS-based fabrication process for vaporizing liquid microthruster silicon device; section 4 presents briefly design and uncertainty analysis of the mN-range hanging pendulum thrust stand; section 5 gives detailed results of thrust measurements; and section 6 concludes the whole work.

2. Design strategy

2.1. Introduction

The schematic diagram and photograph of the current MEMS-based

vaporizing liquid microthruster is shown in Fig. 1. The scope of the current design study includes: 1) vaporization chamber; 2) ohmic heater elements; and 3) converging-diverging nozzle in the vaporizing liquid microthruster architecture. The thrust production of vaporizing liquid microthruster is based on building up of total pressure, accompanied by rapid increase in specific volume of the propellant in the vaporization chamber, increasing the momentum of flow out of the converging-diverging nozzles. The economy of propellant use, as indicated by the specific impulse performance, is dependent on temperature rise within the vaporization chamber.

The starting point in the design is the choice of operating condition. The initial choice is an operating pressure of 2 bar and temperature of 448 K, limited by the thermal stability and structural integrity of the microthruster silicon device assembly. The design propellant flow rate is 5 mg/s at inlet temperature of 293 K with a heater power input of 10 W.

The current microthruster design is equipped with two molybdenum heaters incorporated in the interior of the vaporization chamber for direct heat transfer to the liquid propellant. The two heater elements are capable of providing a peak heat flux of $3.7 \times 10^{-5} \text{ W/m}^2$ with an estimated area averaged heat flux of $1 \times 10^{-5} \text{ W/m}^2$ provided in the vaporization chamber for energy additional to the propellant. The heater elements, submerged in the propellant flow, are glazed in a layer of silicon oxide passivation layer for longevity. These design features enabled at least 50 K superheating of water propellant at the design operating temperature and pressure.

Table 2
MEMS-based vaporizing liquid microthruster comparison: power consumption.

Ref.	$\dot{E}$ [W]	$R_{F-\dot{E}}$ [μN/W]	T_{prop} [K]	T_{in} [K]
[32]	7.8–10.8	19–43	no data	355.5
[26]	0.8	40	no data	none
[34]	4.9 ^a	0.0959 ^a	no data	none
[39]	2.4	50	no data	none
[43]	no data	no data	573	none
[45]	no data	no data	423–573	none
[40]	1.6–3.6	94–280	453	none
[38]	7.1–9.2	4.73–7.29	399–421	none
[47]	7.29–8.76	401.1–428.4	424	none
This work	7.70–7.94	128–256	441–452	294

^a Root mean square value.

Table 3
MEMS-based vaporizing liquid microthruster comparison: configurations.

Ref.	Integral heater element	Heater element material	Heater element passivation	Internal/external heater	Nozzle throat dimensions [$\mu\text{m} \times \mu\text{m}$]
[32]	yes	P-doped Si ^c	none	external	50×50
[26]	yes	gold	none	internal	50×50
[34]	yes	titanium	none	internal	30×30
[39]	yes	B-doped Si ^d	SiO ₂	internal	50×50
[43]	none	none	none	external	120×150
[45]	none	none	none	external	100×100–100×1000
[40]	yes	B-doped Si ^d	none	external	100×130
[38]	yes	printed metal ^e	not specified	internal	200×220
[47]	yes	molybdenum	none	external	20.1×100–26.0×100
This work	yes	molybdenum	SiO ₂	internal	112×40

^c Phosphorus diffused silicon.

^d Boron diffused silicon.

^e Metal not specified.

For the current design the vaporization chamber and the nozzle are fabricated in a single piece on a 100 mm diameter silicon wafer. The flow channel is enclosed by a capping glass wafer which has ports for the propellant feed and electrodes of the heater elements. The heater elements are fabricated on the internal side of the vaporization chamber.

The theoretical background and the governing equations for converging-diverging nozzle and rocket applications are covered in many aerodynamics and thermodynamics textbooks and journal papers (such as [37,47,55,56]). For brevity, this article only discusses the design of MEMS-based vaporizing liquid microthruster based on the following engineering considerations: 1) aerothermal considerations for the choice of vaporization chamber configuration; 2) sizing and material choice for heater element; and 3) silicon wafer fabrication process considerations.

2.2. Aerothermal considerations for vaporization chamber

In early resistojet design such as Marquardt ART-150-A in late 1960's [3,5], the vaporization chamber design has smooth straight channel; in a design variant the vaporization chamber is arranged in a multi-pass configuration with thermal insulation and vacuum jacket to reduce heat loss. In recent design studies on vaporizing liquid microthruster [45–47], various heat transfer enhancement features such as circular and diamond-shaped pedestal arrays and serpentine channels, mainly to maximize the heat transfer area of vaporization chamber for more efficient heat addition, have been examined.

For the current MEMS-based vaporizing liquid microthruster design, smooth straight vaporization chamber configuration was chosen to reduce the loss in the capacity for the propellant to produce work associated with total pressure loss in the vaporization chamber. Furthermore, the loss is more severe at higher temperature. Consider temperature-entropy diagram for pure water vapour, the fact that the isobaric lines diverge as entropy increases indicates that for the same pressure loss the flow would generate more entropy if the pressure loss

occurs at higher temperature during heat addition. Gouy-Stodola theorem [57] states that the lost in power ($\dot{W}_{\text{lost}}$) is directly proportional to the entropy generation rate ($\dot{S}_{\text{gen}}$):

$$\dot{W}_{\text{lost}} = T_{\text{out}} \dot{S}_{\text{gen}} \quad (1)$$

A simple calculation using in equation (1) reveals the sensitivity of loss in resistojet useful power output to total pressure loss in a vaporization chamber: at laboratory condition with room temperature of ~293 K, for a water resistojet that consumes a nominal propellant mass flow rate of 5 mg/s operating at 2 bar absolute pressure and static temperature of 448 K, an additional pressure loss of 5% (0.1 bar) in the resistojet evaporation chamber would lead to an increase in propellant specific entropy of 0.0247 kJ/kg, which corresponds to a loss in available power of 36 mW; for a thrust heater power consumption of 10 W (the current design power input), this loss would represent a 0.4% loss in thermal efficiency. Pedestal arrays and serpentine channel shapes are not implemented in the current design as this would impose further total pressure loss.

Given a propellant mass flow rate of 5 mg/s and heater element power of 10 W, to limit the loss in work to 0.2%, the pressure loss should be limited to ~ 2% of the operating pressure (equivalent to ~4 kPa at 2 bar absolute propellant feed pressure). Using the Darcy-Weisbach equation for flow in the laminar regime to evaluate the Darcy friction factor, the flow velocity limit within the vaporization chamber should be 15 m/s.

Once the heater elements is sized, the footprint of the vaporization chamber can be determined; the depth of vaporization chamber is then designed according to the design flow velocity or other manufacturing constraints.

2.3. Design of heater element

The current vaporizing liquid microthruster is designed to operate in the film boiling regime, in which liquid water propellant vaporizes rapidly before it reaches the walls of the vaporization chamber. With a

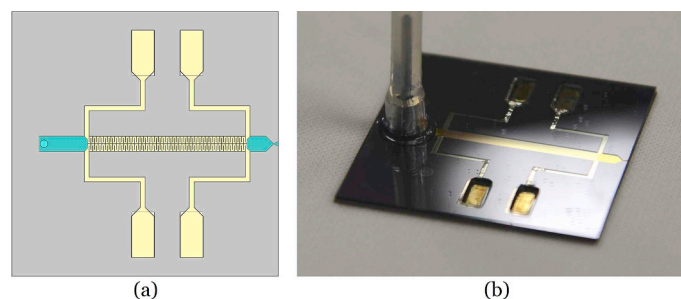


Fig. 1. Schematic and photograph of the current MEMS-based vaporizing liquid microthruster silicon device.

Table 4

Thermal expansion coefficient of materials used in the current microthruster design.

Material	α_{th-exp} [K^{-1}]	Ref.
Silicon	2.5×10^{-6}	[61]
LPCVD silicon nitride	3×10^{-6}	[61]
LPCVD silicon oxide	$0.5 \times 10^{-6} - 4.1 \times 10^{-6}$	[62]
Molybdenum	4.8×10^{-6}	[63]

channel height 200 μm the vaporization chamber may be categorized as microchannel. Oscillatory flow behavior in an elongated microchannel, where the flow moves back-and-forth along the streamwise direction, is known to occur near the critical heat flux boiling regime when the microchannel is mostly filled in water. The cause of this flow oscillation is the compressibility of the rapidly growing vapour bubbles formed in the microchannel [58]. To eliminate this instability, wetting of the vaporization chamber is avoided in the current design. In order to achieve rapid boiling with limited power input, the operating temperature of the microthruster must be held at an excess temperature ($\Delta T_{bx} = T_g - T_{sat}$) greater than that corresponds to the critical heat flux (see the typical boiling curve for water in reference [59]). At an operating pressure of 2 bar, the excess temperature (ΔT_{bx}) at which the critical heat flux boiling occurs is ~ 30 K.

The cohesion of the heater material on the silicon substrate is also critical to implementation of the high heat flux heater elements. Molybdenum is chosen as the heater material as it has a linear thermal expansion coefficient (α_{th-exp}) comparable to that of the silicon wafer substrate (see Table 4). The operation of molybdenum heater has also been proven in references [47,60].

The miniature ohmic heater element is designed with reference to the National Institute of Standards and Technology standard MEMS library [64]. For a compact heater element design with a footprint limited to 2×20 mm, the heater element is fabricated to a sheet resistance of $< 0.2 \Omega/\square$. This is achieved with two molybdenum tracks of ~ 5000 Å thick, 200 μm wide and 67.3 mm long each; the total heat transfer area of the two heater elements is $2.69 \times 10^{-5} m^2$. The design heat flux is $3.7 \times 10^{-5} W/m^2$ on the heater track material and the design heat flux averaged over the outer bound the footprint of the vaporization chamber is $1 \times 10^{-5} W/m^2$ with a design margin of $> 100\%$. This design has achieved an excess temperature (ΔT_{bx}) of > 50 K above the saturation temperature at propellant pressure of 2 bar; thus rapid boiling at reduced heat flux, lower than the critical heat flux, can be achieved.

To prevent the molybdenum tracks from oxidation, the tracks are encased in silicon oxide passivation layer. The effectiveness of the passivation layer is demonstrated in Fig. 2(b). The discolouration of molybdenum tracks associated with the oxidation can be seen on the sample without passivation which is not visible in the sample with passivation.

2.4. Silicon wafer fabrication process considerations

The performance of the silicon-based fabrication processes is an important design consideration for the vaporization chamber depth. The maximum depth of the vaporization chamber is limited to 200 μm to maximize the yield of the deep reactive-ion etching (DRIE) process at the local microelectromechanical system fabrication facility.

The minimum width of the nozzle throat is also constrained by the deep reactive-ion etching process; with the equipment and the recipe available at the local microelectromechanical fabrication facility the 'vertical' walls have a slope of about 85° . In early micronozzle work [65], a supersonic flow nozzle with throat width of 19 μm and expansion ratio of 5.6:1 have been tested; in a recent vaporizing liquid microthruster design study ([47]) a range of throat width of 20.1–26.0

and expansion ratio of 19.5:1–29.9:1 have been tested. After fabrication trials, microthruster samples with the nozzle throat width of 40.2 μm and expansion ratio of 14.2:1 (nozzle exit width = 570 μm) were produced (Fig. 3(b) and (a)). The design here is more conservative where the throat is made slightly wider to reduce the chance of blockage during operation and test.

Another constraint is the accuracy of the heater element patterning in the photolithography process; the edges of the heater element photolithography pattern would blur for vaporization chamber that is too deep. In the current microthruster design the heater element is designed to have smallest feature size of 20 μm , after initial fabrication trials it is found that with a depth of 112 μm the quality of the photolithography pattern remains acceptable.

2.5. Overall dimensions

The aerothermal consideration and MEMS-based fabrication constraints result in a design with a vaporization chamber of ~ 31 mm streamwise length, 2 mm span, and 112 μm depth; heater elements with serpentine molybdenum tracks of overall footprint ~ 20 mm streamwise length, ~ 2 mm span, and ~ 5000 Å thick located on the internal side of the vaporization chamber; nozzle throat width and length of ~ 31 mm and 40 μm respectively. The detailed dimensions are specified in Table 5.

3. Fabrication process flow

The fabrication route was chosen based on the specifications and availability of the equipment for the fabrication process at the local MEMS fabrication facility; the equipment used in the current work or equipment with similar specifications should be commonly used in many other MEMS fabrication facilities. The processes in the MEMS-based fabrication of vaporizing liquid microthruster, with the deposition of each material on the silicon substrate indicated in each step, are shown in Fig. 4.

The fabrication process of vaporizing liquid microthruster starts with a standard single-side polished 100 mm diameter, 525 μm thick silicon wafer (step i in Fig. 4). The first step is to create the flow channel geometry which includes the vaporization chamber and nozzle on the silicon substrate. The Bosch process is carried out for the inductively coupled plasma deep reactive-ion etching (step ii in Fig. 4), ~ 112 μm of material within the boundary imprinted by the photolithography is removed from the silicon substrate.

To reduce the electrical current leakage between the heater element metal tracks, a silicon nitride layer is deposited on the floor of the flow channel (step iii in Fig. 4). A thermal oxide layer is grown on the silicon substrate using wet oxidation before the low-stress silicon nitride is deposited on top of the silicon oxide layer using the low pressure chemical vapour deposition (LPCVD) process; both layers are ~ 1000 Å thick. The silicon nitride, which is non-porous, non-conductive, has high thermal and chemical stability and substantial hardness, serves as the barrier layer, while the thermal oxide layer serves as the interfacial layer improving the adhesion of silicon nitride.

In order to prepare for the bonding of the borosilicate glass capping wafer with the silicon device using anodic bonding (step ix in Fig. 4), the protective layer surface outside the flow channel geometry is removed (unwrapping, step iv in Fig. 4). The surface is then unwrapped using a polysilicon plasma etcher for plasma etching (step iv in Fig. 4).

The molybdenum, ~ 5000 Å thick, is deposited onto the entire device wafer using a magnetron sputtering system (step v in Fig. 4). Wet etching is then carried out using a phosphoric acid based aluminium etchant (step vi in Fig. 4).

A silicon oxide passivation layer, ~ 2000 Å thick, covering the heater elements and the floor of the flow channel is deposited using plasma enhanced chemical vapour deposition (PECVD) at a platen temperature of 573 K (step vii in Fig. 4). The surface outside of the flow

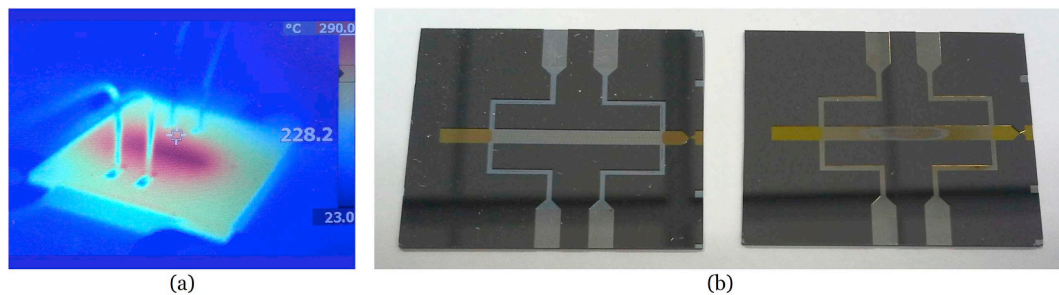


Fig. 2. The current MEMS-based vaporizing liquid microthruster in initial thermal integrity tests, where (a) infrared thermal image, and (b) effect passivation of heater element, left: with passivation, right: without passivation.

channel as well as the areas for heater element electrical contacts are then unwrapped again using wet etching in 777 pad etch, an ammonium fluoride based etchant which etches the silicon oxide without attacking the molybdenum (step *viii* in Fig. 4).

The preparation of the device wafer with all the flow channel cavity, heater elements and the passivation layer is complete up to this step. Anodic bonding is then carried out for the permanent bonded of the silicon device wafer to the boroscillate glass capping wafer which has ports for propellant feed inlet and heater element electrical contacts pre-drilled (step *viii* in Fig. 4); this step encloses the flow channel.

4. Direct thrust measurement stand

4.1. Thrust stand configuration

The mN-range hanging pendulum thrust stand is a fully calibrated direct thrust measurement stand design and built in the Hong Kong University of Science and Technology. The photographs of the facility are shown in Figs. 6 and 8, the schematic diagram of the thrust stand and the peripherals setup including the pressurized propellant feed is shown in Fig. 5. The experimental setup is described with reference to Fig. 5 in the this section.

The design of the thrust stand is kept simple: it consists of a vertical pendulum arm fixed to a crossbeam (item *D*), each side of the crossbeam is simply supported by a pivot fixed to an inertial frame of reference (item *E*); a spiral spring (item *F*) links elastically the motion of the pendulum arm to the inertial frame of reference. The resultant motion of the pendulum arm is the rotation about the axis passing through the two pivots where the amount of pendulum arm rotation (displacement) is directly proportional to the force acting on the pendulum arm. The vaporizing liquid microthruster (item *A*) is mounted directly on the pendulum arm of the thrust stand (Fig. 8(c)), the thrust pushing the pendulum arm is inferred from the displacement (Δx between item *B* and item *C*: Micro-Epsilon Messtechnik capaNC DT sensor head and UNS A96061 aluminium block 12 mm thick and with a width 7 times of the sensor head) measured by a capacitive non-contact displacement measurement system which consists of the sensor head and

Table 5

Dimensions of the current MEMS-based vaporizing liquid microthruster silicon device.

Feature	Size	Unit
Heater element overall length	20180	[μm]
Heater element overall span	1930	[μm]
Heater element thickness	~5000	[$\text{\AA}$]
Heater element metal track width	200	[μm]
Heater element metal track total length	67.3	[mm]
Vacuum chamber overall length	30977.5	[μm]
Vacuum chamber overall width	2000	[μm]
Vacuum chamber depth	112	[μm]
Nozzle throat area	40×112	[$\mu\text{m} \times \mu\text{m}$]
Nozzle exit width	600	[μm]
Nozzle divergent angle	23.28	[$^\circ$]
Propellant feed inlet diameter	1	[mm]
Device overall dimensions	35×35×1.05	[mm×mm×mm]

the controller (items *B* and *V*: Micro-Epsilon Messtechnik capaNC DT CS1 sensor head and 6500 series controller system). The propellant (item *H*: deionized water) and the constant current electrical power supply (item *G*) is fed to the microthruster via thin flexible tube (Tygon tube, outer $\varnothing 2.4$ mm, 0.8 mm wall thickness) and bare copper wires (22 AWG, 2 strands per connection) which are hung from a fixture mounted on the inertial frame of reference and dropped through the opening on the pendulum arm crossbeam (Fig. 8(d)). The use of thin tubing and wiring reduces the interference in the pendulum motion, these attachments are installed such that they the motion of the pendulum arm must not be hindered (Fig. 8(c)). Sufficient distance is allowed between the locations where the feeds are attached to the pendulum arm and inertial frame; this is to reduce the interference of pendulum arm's motion associated with the stiffness of the tubing and wiring.

The thrust stand is housed inside an acrylic vacuum chamber (item *W*) with interior of size 500 mm × 500 mm × 500 mm; acrylic vacuum chamber is chosen for its high visibility and minimal weight. The pressure in the vacuum chamber is measured with a capacitance manometer. To eliminate any vibration transmitted from the floor to

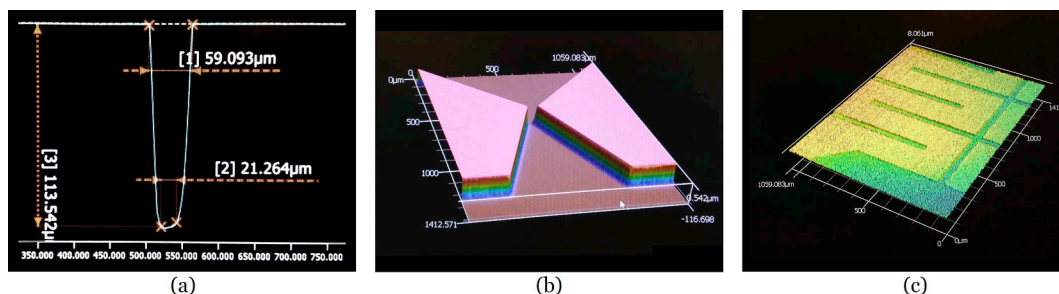


Fig. 3. Metrology for the vaporizing liquid microthruster silicon device; (a) nozzle throat profile, (b) nozzle 3-dimensional profile, (c) heater element 3-dimensional profile.

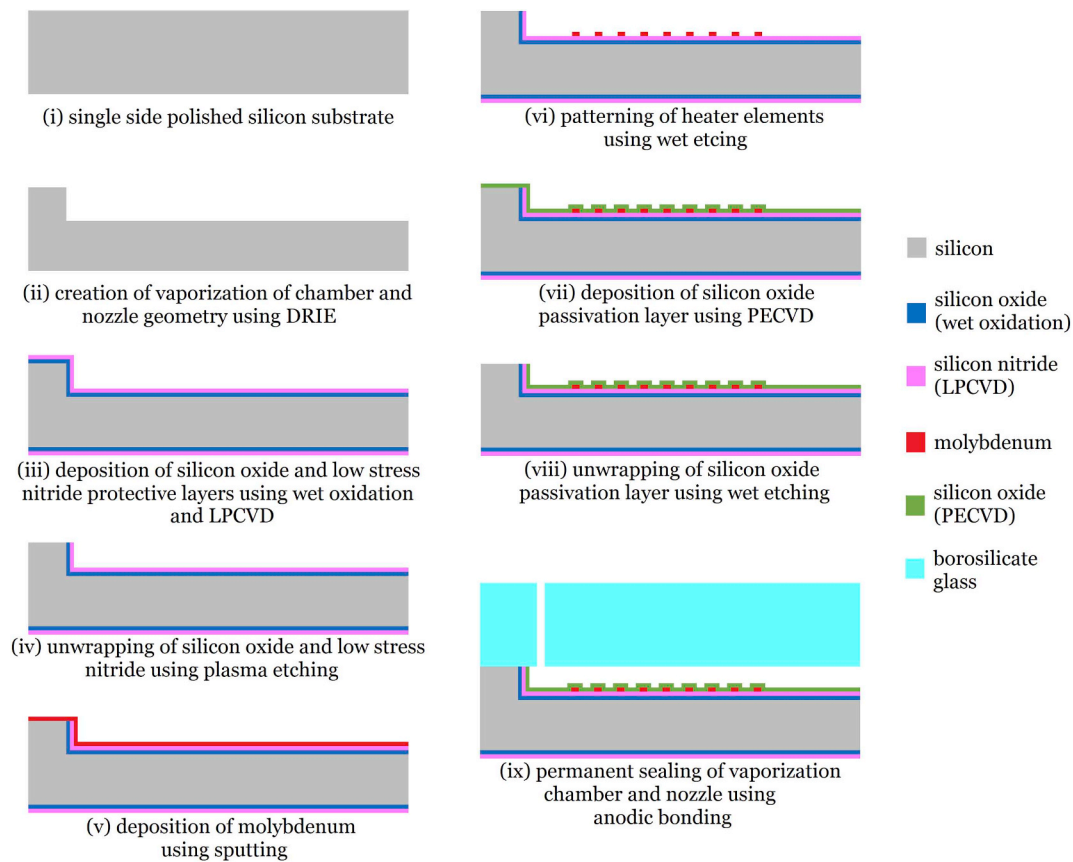


Fig. 4. MEMS-based fabrication process flow for the current vaporizing liquid microthruster.

the pendulum arm, the acrylic vacuum chamber is supported on two pneumatic vibration isolation platforms (item *W*).

The non-contact solenoid force actuator consists of a permanent magnet (item *R*: 5×1 mm neodymium iron boron magnet) and a variable-turn density solenoid coil (item *S*: 16 AWG copper wire with optimized coil density variations). The permanent magnet is held at the

tip of a cantilever attached to the pendulum arm of the thrust stand, energizing the solenoid coil exerts an electromagnetic force directly to the pendulum arm. The solenoid coil of the force actuator is connected to a direct current electrical power supply via a current sensor (item *T*); for accurate current measurement external mechanical ventilation is provided to maintain an acceptable current sensor temperature (below

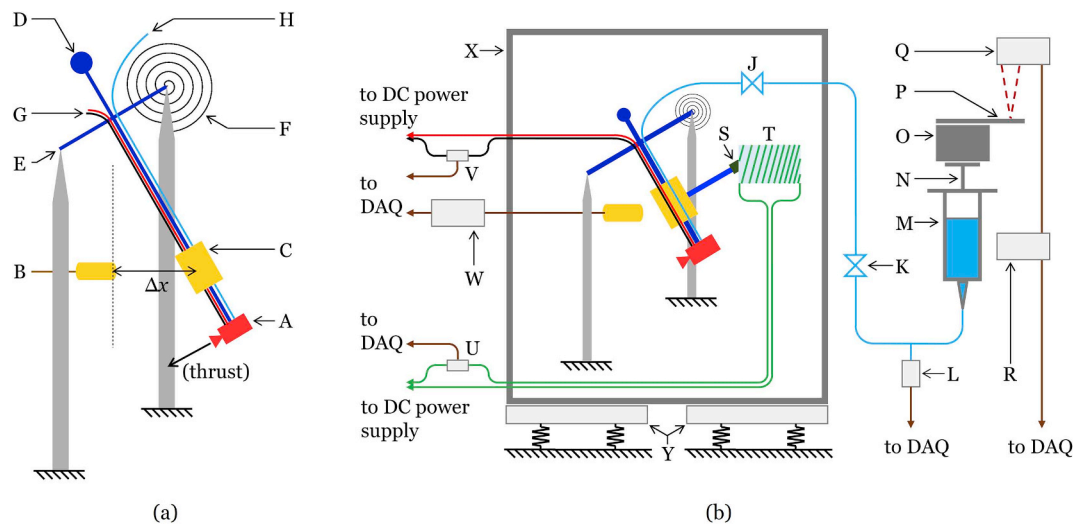


Fig. 5. Schematic of vaporizing liquid microthruster testing facility: (a) thrust stand; (b) overview of experimental setup. Legend: A, vaporizing liquid microthruster; B, capacitive non-contact displacement sensor head; C, capacitive non-contact displacement sensor target plate; D, pendulum arm; E, low-friction pivot; F, spiral spring; G, microthruster electrical feed; H, microthruster propellant feed; I, solenoid isolator valve; J, manual isolator valve; K, propellant feed pressure transducer; L, syringe body; M, syringe plunger; N, deadweight; O, laser non-contact displacement sensor target plate; P, laser non-contact displacement sensor head; Q, laser non-contact displacement sensor controller; R, force actuator permanent magnet; S, force actuator variable-turn density coil; T, current sensor for solenoid for force actuator; U, current sensor for microthruster; V, capacitive non-contact displacement sensor controller; W, vacuum chamber; and X, pneumatic vibration isolator platforms.

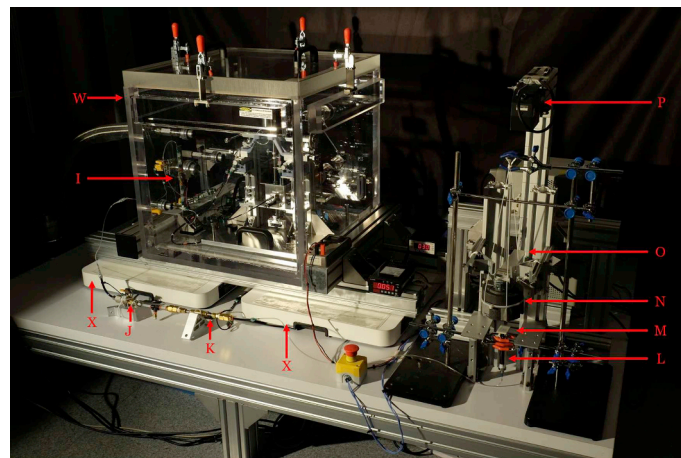


Fig. 6. Photograph of the vaporizing liquid microthruster testing facility at the Hong Kong University of Science and Technology.

343 K the maximum resistance change at this temperature within is 2.2%). The non-contact solenoid force actuator provides a known amount of force directly proportional to the current supplied to the solenoid coil; this is calibrated against an analytical balance before mounting into the thrust stand for pendulum arm force-displacement calibration.

The propellant is fed from a constant pressure syringe pump consists of a vertically mounted glass syringe (item L: 10 mL borosilicate glass syringe) completed with a PTFE-tipped metal plunger (item M) rated at 200 psi, and a set of deadweight (item N). The flow of the propellant is enabled by two isolation valves (items I: solenoid valve and J: manual ball valve). The propellant feed is pressurized by loading the deadweight on top of the plunger. The propellant pressure is measured by an absolute pressure transducer (item K). The propellant volume flow rate is inferred from the velocity of plunger after the propellant pressure is stabilized; the relative position of the plunger (between O and P) is measured by a laser non-contact displacement measurement system which consists of the sensor head and the controller (item P and Q: Keyence LK-503 and LK-2503).

The electrical feed of vaporizing liquid microthruster is connected to a constant direct current power supply via a current sensor (item U) which is also mechanically ventilated. The resistance measurements from the voltage and current of the microthruster is used to infer the temperature of the microthruster device which is assumed to be similar to the propellant temperature; the experimental setup for the temperature-resistance test is shown in Fig. 7.

The signals from all sensing system is fed to a data acquisition system (National Instruments PXIe-1073 chassis completed with a PXIe-4497 module). The following data are recorded: 1) pendulum assembly displacement; 2) voltage and current of vaporizing liquid microthruster;

3) propellant feed pressure; 4) propellant syringe plunger displacement; 5) vacuum chamber pressure; 6) current of non-contact solenoid force actuator.

The thrust produced by the microthruster (F_{thrust}) is related to the time averaged displacement of the pendulum arm ($\Delta\bar{x}$) via the force-displacement calibration factor ($C_{F-\Delta\bar{x}}$):

$$F_{\text{thrust}} = C_{F-\Delta\bar{x}} \Delta\bar{x}. \quad (2)$$

The specific impulse (I_{sp}) is evaluated from the thrust and mass flow rate ($\dot{m}_{\text{prop}}$) measurements:

$$I_{\text{sp}} = \frac{F_{\text{thrust}}}{\dot{m}_{\text{prop}} g_0}. \quad (3)$$

It is worthwhile to emphasize that the same equation has been adopted for resistojet specific impulse calculation [66]. While many research workers [26,32,34,38–40,43,45,47] exclude the energy storage in the scope of their investigation, the mass of energy storage per unit energy for thrust production is considered in this study and an estimate of the thrust and specific impulse produced per mass of energy storage is provided in section 5.4 for completeness.

4.2. Experimental procedure

In a typical experimental run the following steps are carried out:

- i pump test chamber down to ~ 100 Pa;
- ii power up and pre-heated vaporizing liquid microthruster (item A) to the target temperature (463–473 K);
- iii fill the propellant feed syringe (item L) with deionized water;
- iv load deadweight (item N) onto the syringe plunger (item M) for

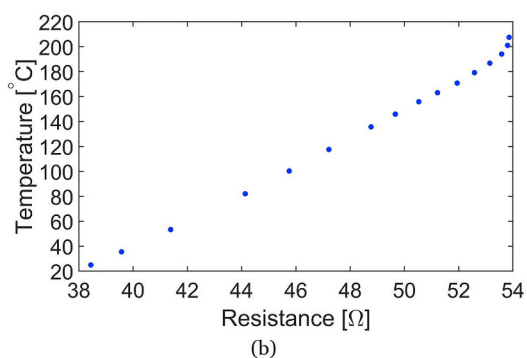
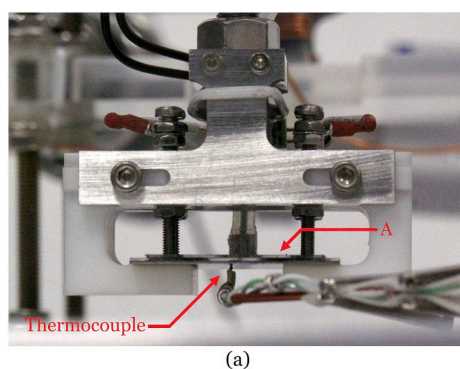


Fig. 7. Temperature-resistance calibration of vaporizing liquid microthruster heater elements.

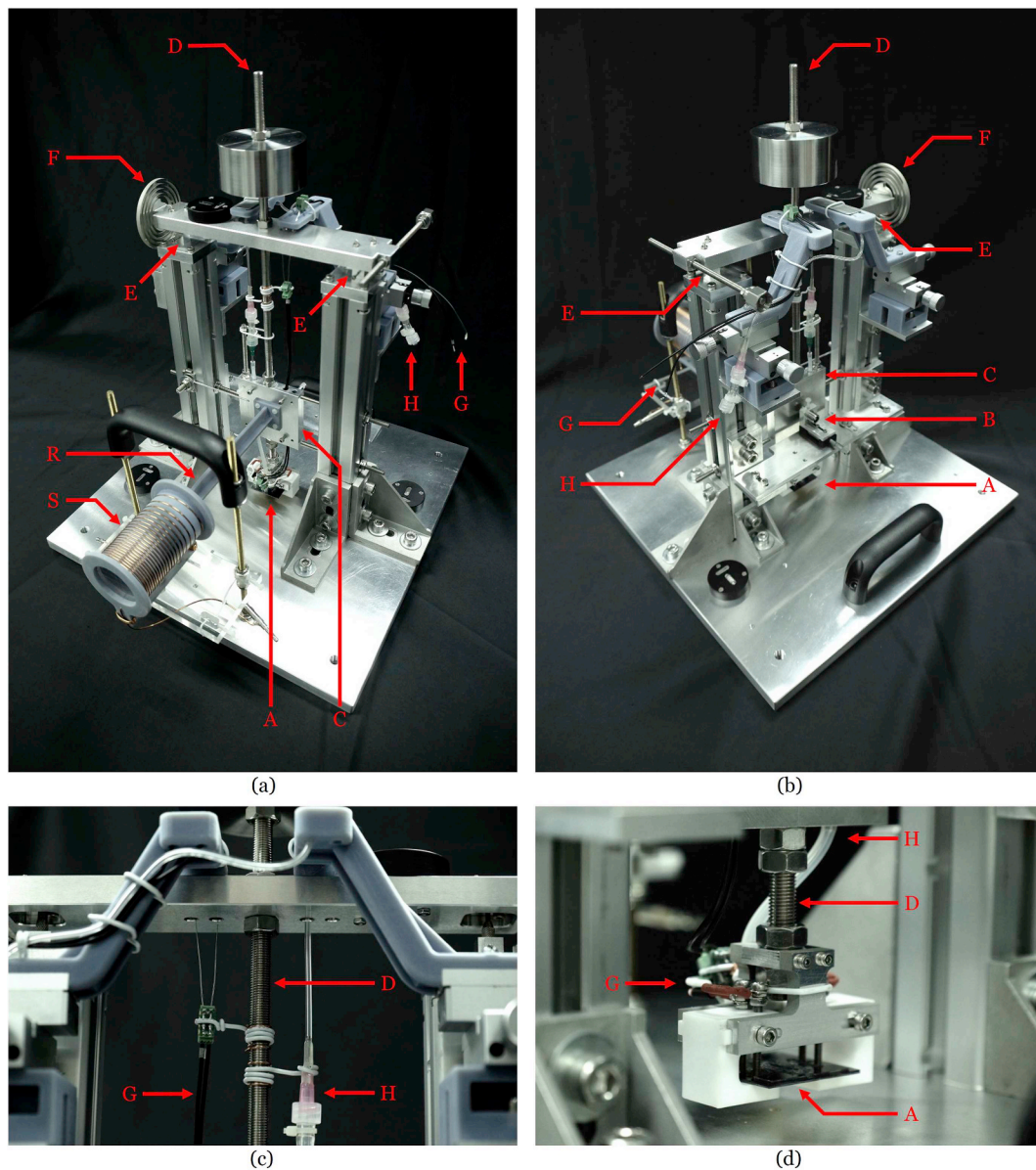


Fig. 8. Photographs of the mN range hanging pendulum thrust stand.

- constant pressure propellant pressurization;
- v wait for all plunger position (item *O*) and pendulum arm (item *D*) to settle;
- vi start data acquisition;
- vii open propellant feed isolation valves (items *I* and *J*) to enable propellant flow;
- viii thrust production begins;
- xi observe pendulum arm displacement and test chamber pressure;
- x shut off propellant feed isolation valves (items *I* and *J*) to cut off thrust;
- xi stop data acquisition and end experiment.

In this series of experiment, the test pressure external to nozzle set in the step *i* is in range 90–100 Pa; this is much higher than the atmospheric pressure at low earth orbit, which ranges between $1.3 \times 10^{-7} - 66.7 \times 10^{-7}$ Pa at ~500 km [67]. The pressure ratio across the vaporizing liquid microthruster nozzle in the current test is much lower than in the orbit, thus thrust performance measured is minimum performance guaranteed for the orbital environment. Step *ii* enables the boiling of liquid water propellant to occur rapidly in the film boiling

regime with an excess temperature above the critical heat flux, this also enable an elevated pressure in entire the propellant feed system to be maintained.

During step *iii* all dead volume of the propellant feed system is filled, a small amount of propellant may be ejected from the vaporizing liquid microthruster in the process and this may perturb the position of underdamped pendulum arm (item *E*) by a small amount before the test. Although the small oscillations sustain for a long time (> 15 minutes), the small oscillations do not cause offset in the averaged pendulum arm displacement as long as propellant is not consumed after the filling and no thrust is produced.

The loading of the dead weight on the plunger of the syringe pump (item *M*) in step *iv*, a constant propellant feed pressure is produced. The propellant flow is considered stable when both the propellant pressure and volume flow rate stabilize, where the volume flow rate is indicated by the rate of change of syringe plunger position (item *O*). The syringe plunger and all the mass attached (items *M*, *N*, and *O*) drop vertically downwards away from the laser displacement sensor (item *P*) at a rate directly proportional to the volume flow rate of the propellant. In the current tests, the settling time is ~20 s after step *vii*; this value is

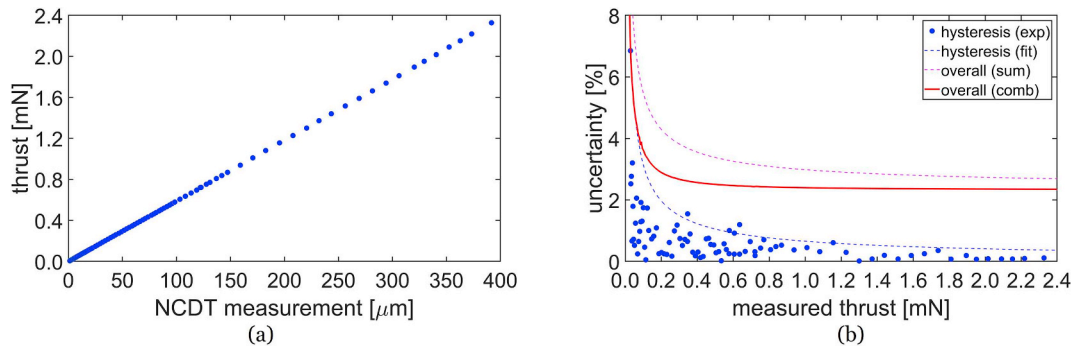


Fig. 9. Calibration of milli-newton hanging pendulum thrust stand: (a) thrust-displacement calibration data, gradient equals C_{PD-T} ; (b) overall measurement uncertainty.

determined by inspection of the test data during data analysis (see Fig. 12 for typical test run data).

The thrust stand is a simple underdamped spring mass system, the time averaged displacement response of the pendulum arm to the thrust is therefore sinusoidal variation. During thrust production (step vii) the pendulum arm (item *D*) is pushed directly by the microthruster (item *A*), the pendulum arm oscillates about a displacement offset ($\Delta\bar{x}$) with larger amplitude; the offset is directly proportional to the thrust as shown in Fig. 9(a).

4.3. Uncertainty

The uncertainty of each transducer is listed in Table 6. The overall uncertainty is characterized as the combined standard uncertainty of uncorrelated input quantities defined in the industrial standard JCGM 100:2008 [54]:

$$u_c = \sqrt{\sum_{i=1}^{N_y} \left(\frac{\partial f}{\partial x_i} \right)^2 u^2(x_i)}, \quad (4)$$

where the uncertainties of propellant pressure, propellant mass flow

Table 6
Uncertainty of each component in the thrust performance measurements.

Uncertainty components			
Measurement	Component	$u(x_i)$	unit
test chamber pressure	Agilent RGC-150	5	[Pa]
propellant pressure (item <i>L</i>)	Honeywell PX3	1730	[Pa]
syringe plunger position (items <i>Q</i> , <i>R</i>)	Keyence LK-503 + LK-2503	10	[μm]
calibration force	Ohaus PAJ1003C	0.1	[mg]
current (current sensor calibration)	Agilent 34411A	0.008	[mA]
force actuator current (item <i>U</i>)	Arcol HS25 1R F	2.2 [§]	[%]
microthruster current (item <i>V</i>)	Arcol HS10 1R F	1.8 [§]	[%]
resistance (current sensor calibration)	Agilent 34411A	1.1	[Ω]
temperature (thermocouple amplifier)	K-type TC + 595 CE	~1	[°C]
temperature (thermocouple reader)	K-type TC + RS pro RS52	~1	[°C]
pendulum arm displacement	Micro-Epsilon capaNCDT	20	[nm]
pendulum arm settings	vernier caliper	0.01	[mm]
pendulum arm hysteresis	pivot	0.61 [§]	[%]
Voltage	NI PXIe-1073 + PXIe-4497	5.96×10^{-7}	[V]
Time	NI PXIe-1073 + PXIe-4497	5×10^{-9}	[s]
Overall uncertainties			
Measurement		$u(x_i)$	unit
heater element resistance (R_{ht})		1.8	[%]
propellant pressure (P_{prop})		5.0	[%]
propellant mass flow rate ($\dot{m}_{prop}$)		2.4	[%]
propellant temperature (T_{prop})		2	[°C]
solenoid force actuator current-force calibration		2.3	[%]
thrust measurement (F_{thrust})		2.4	[%]
specific impulse measurement (I_{sp})		3.5	[%]
power to thrust ratio (R_{P-E})		2.9	[%]

[§] Maximum value within the current test range.

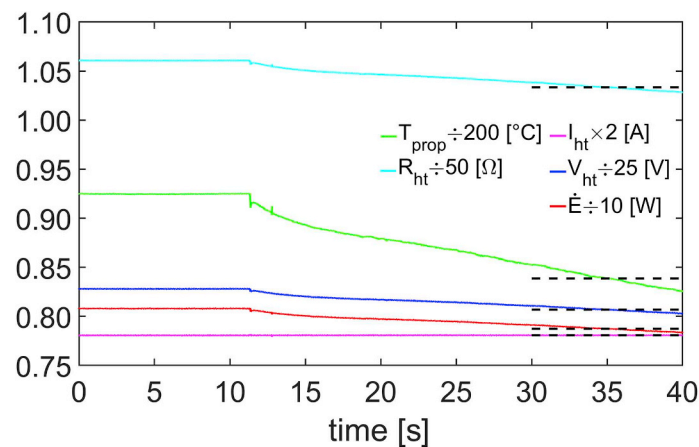


Fig. 10. Transient variation in electrical properties of VLM during operation; the plot has been scaled to show the variations of various parameters clearly.

rate, temperature, voltage and current of the microthruster heater element, solenoid force actuator current and thrust measurements are listed in Table 6.

The main contributors in the force measurement uncertainty are the accuracy of the force-displacement calibration of the solenoid force actuator, measurements of the moment arm, and the hysteresis of the pendulum arm. The overall uncertainty of the thrust measurement is characterized as the combined standard uncertainty (equation (4)) of the hysteresis and the force-displacement calibration evaluated at each calibration data point, its variation with the thrust measurement is shown in Fig. 9(b). At lower force levels the uncertainty is dominated by friction in the pivots. At higher force levels, higher current through the variable-turn density coil (item 5) is required and the uncertainty is dominated by the uncertainty in the current-force calibration of the solenoid force actuator. The main contributor to this uncertainty is the increase in the thermal resistance of current sensor at higher current through solenoid force actuator; the deviation in resistance leads to error in the current measurement.

Inevitably there are some variations in the vaporizing liquid microthruster temperature and test chamber pressure during the tests as the propellant absorbs heat and then subsequently dumped into the test chamber during the thrust measurement; these parameters are time averaged in the data analysis. The range of variations is presented in section 5.3.

5. Thrust measurement and results

5.1. Electrical characteristics

The typical transient electrical characteristics of the vaporizing liquid microthruster powered by constant current source is shown in Fig. 10. Before the propellant flow is enabled, the microthruster is pre-heated to the target temperature by manually increasing the current output until a steady temperature is reached; the power output remains constant at this point as the temperature and thus the resistance of the heater elements remain constant. As soon as the flow of propellant is enabled, the temperature of the heater element is lowered as power is drawn in the rapid vaporization of propellant for thrust generation. This in turn causes the heater element resistance to reduce; note that the heater element resistance is a monotonically increase function of temperature within the range of calibration (see Fig. 7). The transient reduction in heater element resistance should be common to all vaporizing liquid microthruster equipped with metal heater elements. With a reduced heater element resistance, the constant power supply reduces the voltage output to maintain a constant electrical current.

For the following data analysis, the thrust measurements are time-averaged for 10 s after the propellant pressure and mass flow rate

become stable; on average the flow takes ~ 20 s after the propellant flow is enabled. The time-averaged and the change in heater element resistance, voltage, current, and power values over the time period (10 s) which thrust measurements is extracted (indicated by the dashed line in Fig. 10) are extracted are listed in Table 7. The average resistance drop in the whole series of tests is 0.37Ω , which corresponds to a temperature drop of 5 K at a propellant mass flow rate of range $3.06\text{--}5.99$ mg/s. To maintain a constant current output of 0.39 A, the power supply reduces the voltage by 0.16 V (0.8%); this results in reduced microthruster power input of 0.055 W (0.7%).

5.2. Leidenfrost effect in vaporization chamber

The internal flow pattern in the vaporization chamber is captured using a Redlake MotionXtra HG-100K at a frame rate of 5000 Hz. Results are shown in Fig. 11(b)–11(d). The growth of the liquid-vapour interface is shown in Fig. 11(e)–11(f). The Leidenfrost effect in the vaporization chamber is shown in Fig. 11(g)–11(h).

The vaporization chamber is pre-heated to 463 K and is initially dry (Fig. 11(a)). The liquid water propellant turns into mist instantly as soon as the liquid surface reaches the vaporization chamber (Fig. 11(b)). Explosive vaporizing occurs as the pressurized liquid propellant (initially at a pressure is slightly higher than the steady state value) is injected into the vaporization chamber, where the liquid propellant partially vaporized instantly expelling the unvaporized droplets are downstream at a speed of ~ 9 m/s (Fig. 11(b)–11(d)).

In the steady state the propellant pressure settles to a lower value, the mass flow also stabilizes to the rate at which all liquid propellant vaporizes and all droplets are eliminated. The phase interface is well defined where a liquid core is visible in Fig. 11(e) and (f). At an elevated temperature the phase interface contracts further upstream as the liquid propellant takes a less streamwise length for complete vaporization. Note that the phase interface is formed at a distance away from the internal walls of the vaporization chamber consistently, which

Table 7

Electrical parameters of the current microthruster design, averaged values of all tests.

Parameters	Time-averaged value during thrust measurements	Change during thrust measurements	Unit
Current (I_{ht})	0.39	3.47×10^{-5}	[A]
Voltage (V_{ht})	20.22	0.16	[V]
Resistance (R_{ht})	52.09	0.37	[Ω]
Power ($\dot{E}$)	7.84	0.055	[W]
Temperature (T_{prop})	446	5	[K]

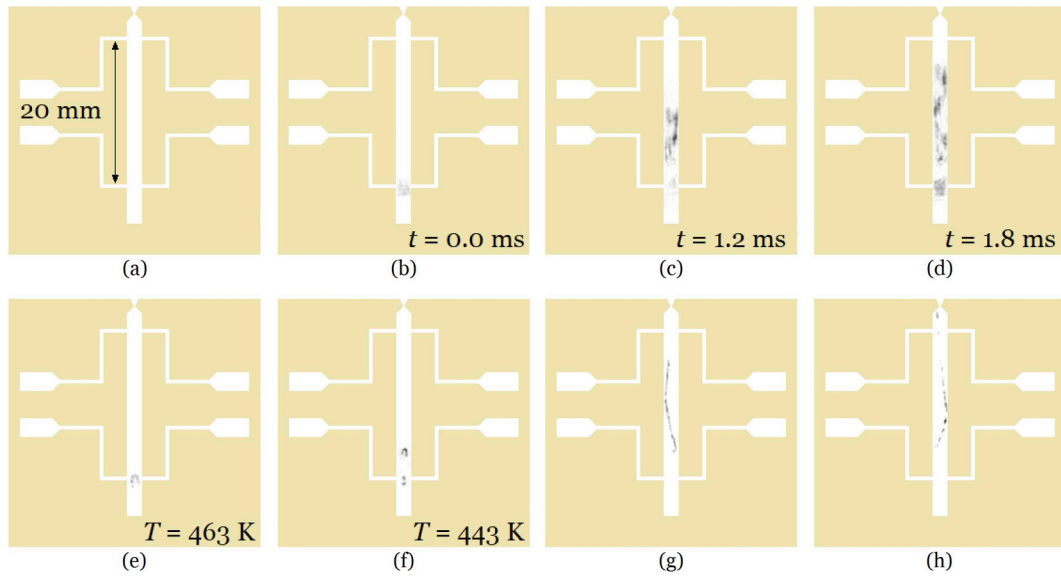


Fig. 11. High speed camera images of internal flow captured at 5000 Hz (the liquid surface and droplets are shown in grey): (a) geometry of a dry vaporization chamber; (b)–(d) initial explosive vaporization at 463 K; (e)–(f) phase interface at different temperature; (g)–(h) trajectory of droplets.

shows that the boiling occurs in the film boiling regime.

Leidenfrost effect, where a layer of evaporated vapour would form between the liquid and the hot boundary, is evident for our design in Fig. 11(g) and (h) which show water droplets bouncing off the hydrophilic silicon dioxide internal surfaces as they are transported in towards the nozzle. The vaporization rate is sufficiently high that the mass transferred across the phase interface of the droplets produces sufficient pressure to repel the droplet away from the internal surfaces of the vaporization chamber. This is contrasted with adhesive behaviour of liquid propellant droplets that would occur at lower boiling excess temperature where the droplets would spread out on the internal hydrophilic surface. The formation of liquid propellant layer due to adhesion and further coalescence of propellant droplets would hinder the flow in the vaporization chamber, which in turn would result in additional losses in total pressure in the vaporization chamber and thus efficiency. The coalescence of propellant droplets is prevented in the current microthruster due to the Leidenfrost effect. To the best of our knowledge, the current design is the first MEMS-based vaporizing liquid microthruster to be proven to achieve continuous thrust production in the film boiling regime with instant vaporization and Leidenfrost effect. The instant boiling is the result of the successful implementation of the high heat flux elements inside the vaporization chamber. This leads to one of the highest specific impulse amongst the MEMS-based vaporizing liquid microthrusters which are rigorously tested using direct thrust measurement methods.

5.3. Transient response of the experiment setup

A series of thrust measurement tests have been conducted at constant power input ($\dot{E}$) of 7.84 W (+0.10/−0.14 W) with propellant feed pressure (P_{prop}) being the main variable between range 1.25–2.22 bar; resulting in output thrust (F_{thrust}) of 1.00–2.00 mN and specific impulse (I_{sp}) of 25.9–51.1 s at a propellant consumption rate ($\dot{m}_{\text{prop}}$) of 3.06–5.99 mg/s and exhaust temperature (T_{prop}) of 446 ± 6 K). The time series measurements in a typical test are plotted in Fig. 12. The thrust (F_{thrust}), specific impulse (I_{sp}), thrust-to-power ratio ($R_{F-\dot{E}}$), and propellant mass flow rate ($\dot{m}_{\text{prop}}$) and propellant feed pressure (P_{prop}) and exhaust temperature (T_{prop}) are measured. The results of the propulsion performance tests are presented in Fig. 13 and Table 8, which include:

- thrust (F_{thrust}) variations with propellant feed pressure (P_{prop});
- propellant mass flow rate ($\dot{m}_{\text{prop}}$) variations with propellant feed pressure (P_{prop});
- specific impulse (I_{sp}) variations with propellant feed pressure (P_{prop});
- specific impulse (I_{sp}) variations with propellant exhaust temperature (T_{prop});
- thrust-to-power ratio ($R_{F-\dot{E}}$) variations with propellant feed pressure (P_{prop}).

With reference to the components of the thrust stand presented in Figs. 5 and 8, the transient response of the thrust stand is described

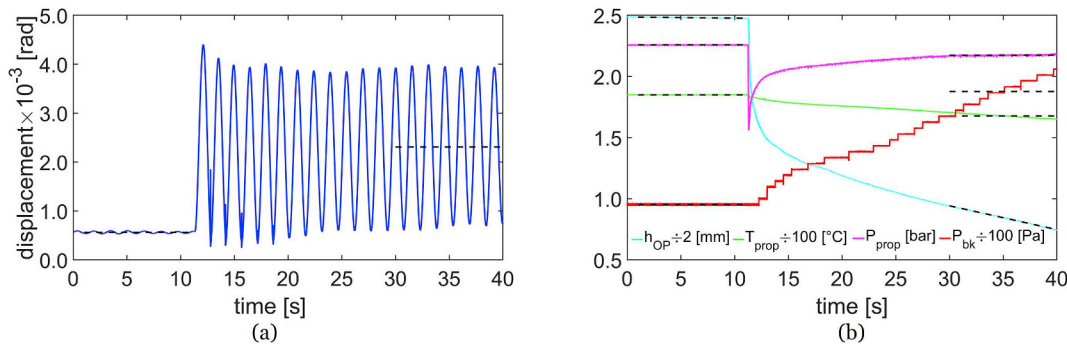


Fig. 12. Typical time series data of thrust performance measurement: (a) pendulum arm displacement; (b) syringe pump plunger position (h_{OP}), propellant temperature (T_{prop}), propellant pressure (P_{prop}), and test chamber pressure (P_{TCh}). Note that the plot has been scaled to show the variations of various parameters clearly.

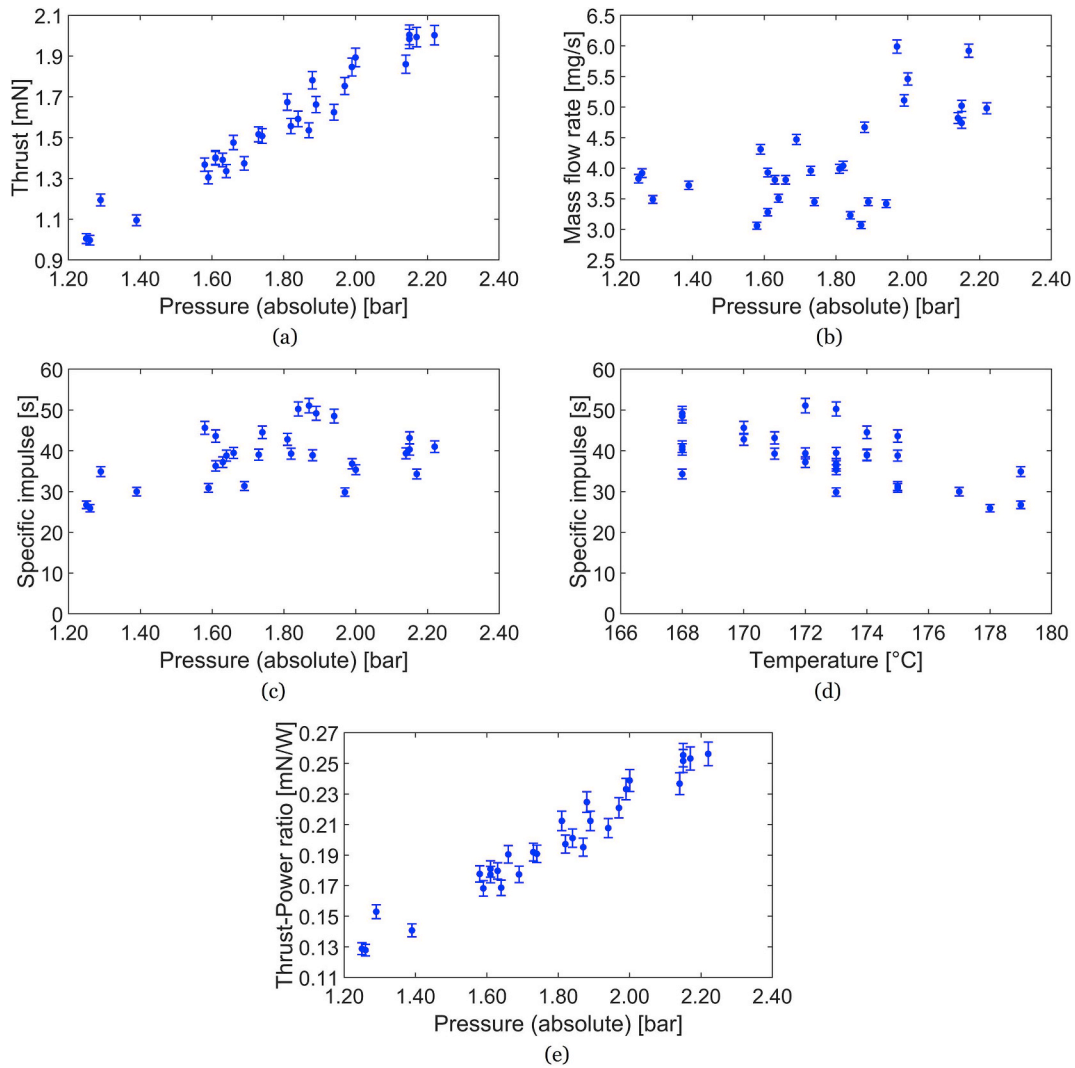


Fig. 13. Summary of thrust performance measurement: (a) thrust (F_{thrust}) variations with propellant feed pressure (P_{prop}); (b) propellant mass flow rate ($\dot{m}_{\text{prop}}$) variations with propellant exhaust temperature (T_{prop}); (c) specific impulse (I_{sp}) variations with propellant feed pressure (P_{prop}); (d) specific impulse (I_{sp}) variations with propellant exhaust temperature (T_{prop}); and (e) thrust-to-power ratio ($R_{\text{F-P}}$) variations with propellant feed pressure (P_{prop}).

here. For all tests in this design study, the microthruster is pre-heated to a temperature of 463–473 K (± 1 K) for at least 10 min and the test chamber is pumped down to 90–100 Pa absolute with propellant flow disabled. The data acquisition starts 10 s before the propellant flow isolation valve (item *J*) is opened. As soon as the propellant flow is enabled the propellant feed tubing is filled, the syringe pump plunger (item *M*) begins to sink as propellant is being expelled from the microthruster for thrust generation and the test chamber begins to rise as the propellant is being ejected into the test chamber. At this point the pendulum arm (item *D*) of the thrust stand is pushed and starts to oscillate about a displacement offset from the initial position in a manner shown in Fig. 12(a), where the offset ($\Delta\bar{x}$) is directly proportional to the averaged thrust level (F_{thrust}).

The thrust generation is considered stable when the propellant feed pressure (P_{prop}) and the velocity of the syringe pump plunger (item *M*) both reach a constant value. Note that the velocity of the syringe pump plunger (item *M*), which is directly proportional to the propellant mass flow rate, is indicated by the rate of change of the distance between the laser non-contact displacement sensor head (item *O*) and target plate (item *P*) (h_{OP}): i.e. the slope of h_{OP} shown in Fig. 12(b); a constant slope indicates a steady value is reached. The settling time is dependent on the volume in the propellant feed tubing, the typically settling time is ~ 20 s in the current experimental setup (as determined from

Fig. 12(b)); there are potentials in further reducing the settling time in a more compact installation. The magnitude of thrust is averaged during the steady state; the time period of the averaging is indicated by a dashed line in Fig. 12.

During thrust generation, propellant continues to be dumped into the test chamber (item *W*, interior of size 500 mm $\times$ 500 mm $\times$ 500 mm). The maximum pressure rise in the test chamber (ΔP_{TCh}) can be estimated using equation of state:

$$\Delta P_{\text{TCh}} = \frac{m_{\text{TCh}} \times R_{\text{H}_2\text{O}} \times T_{\text{TCh}}}{V_{\text{TCh}}}, \quad (5)$$

where m_{TCh} is the mass of propellant accumulated in the test chamber during the averaging time period, $R_{\text{H}_2\text{O}}$ is the specific gas constant of water vapour, T_{TCh} is the temperature in the test chamber which is assumed to be similar to the ambient temperature, and V_{TCh} is the volume of the test chamber. In this test campaign, mass of propellant vapour accumulated in the test chamber (m_{TCh}) during the thrust measurement averaging time period (10 s) ranges between 30.6–59.9 mg, giving a maximum pressure rise (ΔP_{TCh}) of 32.2–62.0 Pa. All through each thrust measurement test the vacuum pump is kept running continuously, the propellant vapour built up in the test chamber is partially removed, resulting in an actual pressure rise of 8.7–34.0 Pa. The rise in pressure due to the accumulation propellant is inevitable; however, the build up

Table 8
Summary of vaporizing liquid microthruster thrust performance tests.

Test	P_{prop} [bar]	$\dot{E}$ [W]	$\dot{m}_{\text{prop}}$ [mg/s]	F_{thr} [mN]	I_{sp} [s]	T_{prop} [°C]	$R_{F-\dot{E}}$ [mN/W]
1	1.25	7.81	3.83	1.005	26.7	179	0.129
2	1.26	7.80	3.92	0.997	25.9	178	0.128
3	1.29	7.81	3.49	1.194	34.9	179	0.153
4	1.39	7.78	3.72	1.095	30.0	177	0.141
5	1.58	7.70	3.06	1.368	45.6	170	0.178
6	1.59	7.76	4.31	1.305	30.9	175	0.168
7	1.61	7.73	3.93	1.399	36.3	173	0.181
8	1.61	7.91	3.28	1.402	43.6	175	0.177
9	1.63	7.74	3.81	1.391	37.2	172	0.180
10	1.64	7.92	3.51	1.336	38.8	175	0.169
11	1.66	7.75	3.81	1.476	39.5	173	0.190
12	1.69	7.74	4.47	1.374	31.3	175	0.177
13	1.73	7.90	3.96	1.516	39.0	174	0.192
14	1.74	7.90	3.45	1.508	44.5	174	0.191
15	1.81	7.88	3.99	1.674	42.8	170	0.212
16	1.82	7.89	4.04	1.557	39.3	171	0.197
17	1.84	7.91	3.23	1.592	50.2	173	0.201
18	1.87	7.87	3.07	1.536	51.1	172	0.195
19	1.88	7.93	4.67	1.782	38.9	174	0.225
20	1.89	7.83	3.45	1.662	49.2	168	0.212
21	1.94	7.82	3.42	1.625	48.5	168	0.208
22	1.97	7.94	5.99	1.753	29.9	173	0.221
23	1.99	7.92	5.11	1.847	36.8	173	0.233
24	2.00	7.93	5.46	1.893	35.3	173	0.239
25	2.14	7.86	4.82	1.860	39.4	172	0.237
26	2.15	7.88	5.02	1.984	40.3	168	0.252
27	2.15	7.85	4.74	2.004	43.1	171	0.255
28	2.17	7.87	5.92	1.993	34.3	168	0.253
29	2.22	7.82	4.98	2.002	41.0	168	0.256

of propellant in the test chamber may be reduced with the use of larger vacuum chamber and vacuum pump of higher pumping speed.

5.4. Propulsion performance

The results presented in Fig. 13 and Table 8 are discussed in this section. Comparing the variations in thrust (F_{thrust}) and specific impulse (I_{sp}) with propellant feed pressure (P_{prop}) in Fig. 13(a) and (c), it can be observed that the thrust is a strong function of propellant feed pressure whereas the specific impulse is not correlated strongly with the propellant feed pressure. It can also be observed that the propellant mass flow rate ($\dot{m}_{\text{prop}}$) also increases with the propellant feed pressure as shown in Fig. 13(b); therefore the increase in thrust can also be correlated to the increase in propellant mass flow rate.

At a constant power input ($\dot{E} \sim 7.84$ W), the thrust-to-power ratio ($R_{F-\dot{E}}$) increases with the propellant feed pressure, as shown in Fig. 13(e). This is consistent with the trend shown in Fig. 13(a).

The efficiency of the use of propellant in thrust production is indicated by the specific impulse ($I_{\text{sp}} = F_{\text{thrust}}/(\dot{m}_{\text{prop}} g_0)$) which is directly proportional to the amount of thrust produced per unit propellant mass flow rate. Within a relatively small exhaust temperature variation, $T_{\text{prop}} = 446 \pm 6$ K, the specific impulse remains invariant with both propellant feed pressure and exhaust temperature, shown in Fig. 13(c) and (d). This suggests all propellant is vaporized within the vaporization chamber at power consumption ($\dot{E}$) as low as ~ 7.84 W and our design is capable of maintaining the efficiency in use the propellant for thrust production over the range of mass flow rate and propellant feed pressure tested.

For a fair comparison, the mass of the energy storage is also considered here. The current MEMS-based thruster head is designed to work with contemporary CubeSat battery technology, mostly based on 18650 Lithium-ion rechargeable battery technology, which has a typical energy capacity of 45–100 Wh, with a mass of 360–700 g and mass-per-energy ratio of 1.8–2.2 g/kJ (such as [68–70]). Allowing an overall electrical power budget of 20% for thrust production, the

equivalent thrust-to-dry-mass ratio and specific-impulse-to-dry-mass ratio for the propulsion system including the energy source (20% of overall battery mass ~ 150 g) and propellant control valve (~ 100 g) is estimated to be 4.0–8.0 mN/kg and 103.6–204.0 s/kg.

The specific impulse results show this MEMS-based vaporizing liquid microthruster is capable of producing a specific impulse level that is the highest amongst other MEMS-based vaporizing liquid microthruster tested rigorous with direct thrust measurement method to-date. The current MEMS-based vaporizing liquid microthruster design also remains a highly competitive option compared with other MEMS-based resistojets such as free-molecular micro-resistojets (such as [48–50]). In addition, the current MEMS-based vaporizing liquid microthruster design offers significant advantage over other contemporary all-metal small-scale resistojets thruster head design (such as [51–53]) with efficient thrust production at a fraction of electrical current.

6. Conclusion

In this work, a MEMS-based vaporizing liquid microthruster is designed, fabricated, and tested using the direct thrust measurement method in the hanging pendulum thrust stand designed from scratch. The unique features of our vaporizing liquid microthruster design is the high heat flux molybdenum passivated in silicon oxide located inside the vaporization chamber rather than external to the vaporization chamber for direct heat addition for vaporization and thus better efficiency; a local heat flux 3.7×10^{-5} W/m² has been achieved. The bespoke direct thrust measurement stand has achieved a hysteresis as low as 0.61% over at the thrust measurement range 1.00–2.00 mN, leading to an uncertainty of 2.4% for thrust measurements and an uncertainty of 3.5% for specific impulse measurements. The microthruster designed, fabricated and tested in this campaign has achieved a continuous thrust level of 1.00–2.00 mN at specific impulse of 25.9–51.1 s with a power consumption of 7.70–7.94 W and propellant mass flow rate of 3.06–5.99 mg/s, operating at propellant feed pressure of 1.25–2.22 bar absolute and temperature of 441–452 K. For an electrical power budget of 20%, the equivalent thrust and specific impulse per device dry mass (including the battery and associated valve) are estimated to be 4.0–8.0 mN/kg and 103.6–204.0 s/kg respectively. This MEMS-based vaporizing liquid microthruster offers a specific impulse level that is the highest amongst the MEMS-based vaporizing liquid microthrusters tested rigorously with direct thrust measurement method to-date; the current design also offers a specific impulse level that is comparable to a contemporary small-scale resistojets design at a fraction of the electrical current.

Future work on MEMS-based vaporizing liquid microthruster design may include a parametric study on planar micro-converging-diverging-nozzles and the effects of gaseous solute in liquid water propellant for enhanced thrust production. Additional flow instabilities may arise from the release of dissolved gases during boiling. Hence, the effect of degassing shall be characterized in further tests. The density of exhaust plume from this vaporizing liquid microthruster should also be characterized to investigate the feasibility of the generation of induction plasma downstream to the vaporizing liquid microthruster nozzle based on pulsed-plasma technologies. Last but not least, successful ignition would enable specific impulse enhancement or thrust vectoring.

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